

DD: SUB-WF-MIGRATION-01 — Subscription Migration

Developer implementation guide:

DD_SUB-WF-MIGRATION-01-Subscription_Migration-DEV_IMPL_GUIDE-v1.1.md.

Coherence note (2026-06-01): cross-technology migration is no longer a vague future WO item. SUB-WF-MIGRATION-01 owns the business operation and subscription state transition. FUL-10 owns the technical cross-technology fulfillment sequence and calls WO/OSR/PROV-INT as needed. Any older "not yet supported" wording must be read as pre-FUL-10 history, not current implementation guidance.

1. Purpose

This rewritten DD is the developer-facing version of the module contract

DD_SUB-WF-MIGRATION-01-Subscription_Migration-v1.1.md.

Verdict: ■ Updated | **Wave:** 1 | **Group I:** Subscription Management | **Version:** v1.1 Operation that transitions an ACTIVE subscription's homepass_id AND package_ref simultaneously. The compound operation distinct from RELOCATION (HomePass only) and from UPGRADE/DOWNGRADE (Package only). Used when: (a) customer requests move AND package change at same time (typical case); (b) operator-driven technology migration (cross-tech retirement cohort, for example HFC -> GPON). Owns: the 2-trigger taxonomy, target-Package + target-HomePass validation, equipmentPath taxonomy, disconnectMode taxonomy, the cycleAnchorPolicy + effectiveTiming taxonomies (since Package change is involved), the integration surface to WO-01 (SHIFTING flow for same-tech case), the per-operator config catalog, the trigger + cancel REST endpoints, the rich domain Kafka events. **Physical work delegated to WO-01/FUL-10 as appropriate.** For same-technology migration (source network root type == target network root type), the workflow creates ONE Work Order with kind=SHIFTING via POST /api/work-orders — same integration as RELOCATION-01 v1.1. The WO module's SHIFTING flow handles physical work internally. Per DD_WO-01-CONTRACT-FOR-SUB-WF. For cross-technology migration, ADMIN_TECH_MIGRATION starts the subscription operation and calls FUL-10 for the technical fulfillment sequence. **Atomic-switch service continuity:** subscription stays ACTIVE on source HomePass + source Package until target activate succeeds. PENDING_MIGRATION is entered only during the commit window. **What changed from v1.0:** - Replaced the made-up "two-WO orchestration" with single SHIFTING WO per DD_WO-01-CONTRACT-FOR-SUB-WF for same-tech cases. - Cross-tech now delegates to FUL-10 instead of a vague future WO satellite. - Added MODIFY_SERVICE step — Package change requires per-Service-class fan-out at target. After target activation, workflow calls FUL-03 v1.1 MODIFY_SERVICE to diff source vs target Package compositions and apply add/drop/modify per service class. - **Operator-neutral framing.** Per-operator concerns (job_type_codes, BPMN file names, equipment recovery policy, account-status codes) live in WO/FUL-10 operator catalogs.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/04_subscription_workflows/DD_SUB-WF-MIGRATION-01-Subscription_Migration-v1.1.md

Item	Value
Version	v1.1
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	rules.subscription.migration.<operator>

4. Runtime Contracts To Implement

4.1 APIs

- DELETE /api/subscriptions/{subscription_id}/pending-migration/{operation_id}
- DELETE /pending-migration/{operationId}
- POST /api/subscriptions/{subscription_id}/migrate
- POST /api/work-orders

4.2 Tables

- subscription_migration_config

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- bil01-emit-intent
- ful-call-activate
- ful-call-claim-homepass
- ful-call-modify-service
- ful-call-release-homepass
- ful-call-suspend
- notification-send
- payment-confirmed
- wo-cancel
- ful10-start-technology-migration
- wo-create-shifting

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- SubscriptionMigrated
- SubscriptionMigrationCancelled

- SubscriptionMigrationRejected
- SubscriptionMigrationScheduled
- WorkOrderCancelled
- WorkOrderFinalized
- WorkOrderTimedOut

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ Updated | **Wave:** 1 | **Group I:** Subscription Management | **Version:** v1.1

Operation that transitions an ACTIVE subscription's homepass_id AND package_ref simultaneously. The compound operation distinct from RELOCATION (HomePass only) and from UPGRADE/DOWNGRADE (Package only). Used when: (a) customer requests move AND package change at same time (typical case); (b) operator-driven technology migration (cross-tech retirement cohort — e.g., HFC → GPON). Owns: the 2-trigger taxonomy, target-Package + target-HomePass validation, equipmentPath taxonomy, disconnectMode taxonomy, the cycleAnchorPolicy + effectiveTiming taxonomies (since Package change is involved), the integration surface to WO-01 (SHIFTING flow for same-tech case), the

per-operator config catalog, the trigger + cancel REST endpoints, the rich domain Kafka events.

Physical work delegated to WO-01. For same-technology migration (source network root type == target network root type), the workflow creates ONE Work Order with kind=SHIFTING via POST /api/work-orders — same integration as RELOCATION-01 v1.1. The WO module's SHIFTING flow handles physical work internally. Per DD_WO-01-CONTRACT-FOR-SUB-WF.

Cross-tech migration delegated to FUL-10. ADMIN_TECH_MIGRATION trigger involves moving customers across network technology (e.g., HFC -> GPON). SUB-WF-MIGRATION-01 validates the business request and owns the subscription operation. FUL-10 owns the technical fulfillment sequence and calls WO-01, OSR, FUL-09, and PROV-INT as required. If FUL-10 is not enabled for the operator, return 400 TECHNOLOGY_MIGRATION_NOT_ENABLED.

Atomic-switch service continuity: subscription stays ACTIVE on source HomePass + source Package until target activate succeeds. PENDING_MIGRATION is entered only during the commit window.

What changed from v1.0:

- **Replaced the made-up "two-WO orchestration" with single SHIFTING WO** per DD_WO-01-CONTRACT-FOR-SUB-WF — same correction as RELOCATION-01 v1.1.
- **Replaced fabricated message catches** (disconnection-completed, installation-completed) with WorkOrderFinalized. Replaced fabricated config columns (disconnect_wo_timeout_days, install_wo_timeout_days) with WO-01's wo_kind_config.default_timeout_days. Single worker wo-create-shifting.
- **Added MODIFY_SERVICE step** — Package change requires per-Service-class fan-out at target. After target activation, workflow calls FUL-03 v1.1 MODIFY_SERVICE to diff source vs target Package compositions and apply add/drop/modify per service class.
- ADMIN_TECH_MIGRATION **delegated** — calls FUL-10 when subscription_migration_config.admin_tech_migration_enabled = true; otherwise returns 400 TECHNOLOGY_MIGRATION_NOT_ENABLED.
- **Operator-neutral framing.** Per-operator concerns (job_type_codes, BPMN file names, equipment recovery policy, account-status codes) live in WO-01 FLOW satellites + operator catalogs.

Glossary

Term	Meaning
Migration	Change of subscription's homepass_id AND package_ref simultaneously. Distinct from RELOCATION (HomePass only) and UPGRADE/DOWNGRADE (Package only).
Source HomePass + source Package	Current snapshots at migration time.
Target HomePass + target Package	Where the subscription is moving and what new Package it will be on. Target Package's required service families MUST be supported by target HomePass.
equipmentPath	KEEP_EXISTING (re-home existing equipment at target — only valid if target Package's equipment requirements match source) or NEW_INSTALL (install fresh equipment, recover old). Auto-inferred + can be overridden.
disconnectMode	TECH_DISPATCH (workflow creates SHIFTING WO) or CUSTOMER_SELF_MOVE (no WO; customer self-handles). NEW_INSTALL always forces TECH_DISPATCH.
cycleAnchorPolicy	Per BIL-01 v1.1 §3.2: PRESERVE (current cycle continues, prorate delta) or RESET_TO_OPERATION_DATE (issue OLD_CYCLE_CREDIT + FIRST_CYCLE_CHARGE). Operator config default; per-request override.
effectiveTiming	IMMEDIATE / END_OF_CURRENT_CYCLE / SCHEDULED_AT. END_OF_CURRENT_CYCLE + PRESERVE short-circuits proration delta to zero (cycle finishes on source, next starts on target).
Same-tech migration	sourceHomePass.networkRootType == targetHomePass.networkRootType. Uses WO-01 SHIFTING flow. Supported in v1.1.
Cross-tech migration	sourceHomePass.networkRootType != targetHomePass.networkRootType. Supported through ADMIN_TECH_MIGRATION when FUL-10 is enabled for the operator. Customer-requested cross-tech remains blocked unless the operator explicitly enables it.

Term	Meaning
SHIFTING WO	A Work Order with kind=SHIFTING per DD_WO-01-CONTRACT-FOR-SUB-WF.

1. What this is

The migration operation. Triggered by: a customer requesting a HomePass change + Package change in one operation (CUSTOMER_REQUESTED_MIGRATION), or admin-driven tech cohort migration (ADMIN_TECH_MIGRATION). Admin-driven cross-technology migration requires FUL-10 to be enabled for the operator.

Outcomes (at actual commit time): subscription homepass_id updated to target, previous_homepass_id snapshot retained, package_ref + package_version_id updated to target Package, previous_package_ref + previous_package_version_id snapshot retained, migration fee + signed proration delta (or RESET cycle pair) charged per BIL-01 v1.1, physical work via WO-01 SHIFTING flow (or self-move), FUL-07 bookkeeping updated on both HomePasses, source HomePass network-side de-provisioned + target HomePass provisioned for target Package via FUL-03 + FUL-04, last_migrated_at master field updated, customer notified.

For non-IMMEDIATE effectiveTiming, the subscription remains ACTIVE on source HomePass + source Package during the wait. A second pending-future migration is rejected; any state-affecting operation during the wait cancels the pending migration.

What this DD owns

1. **subscription_migration_config table** — per-operator config.
2. **Trigger taxonomy** — 2 triggers with role gating.
3. **Target-Package validation rules** — target Package must differ from source, status sellable.
4. **Target-HomePass validation rules** — sellable, status active, service-family compatibility for TARGET Package, port-provisioned.
5. **equipmentPath taxonomy + auto-inference rules** — taking target Package's equipment requirements into account.
6. **disconnectMode taxonomy + resolution rules.**
7. **cycleAnchorPolicy taxonomy + resolution rules.**
8. **effectiveTiming taxonomy + non-IMMEDIATE timer + cancellation paths.**
9. **WO-01 SHIFTING integration sequencing** — same as RELOCATION-01 v1.1, with extra MODIFY_SERVICE step post-activation for Package fan-out.
10. **Trigger + cancel REST endpoints.**
11. **Domain Kafka events** — SubscriptionMigrationScheduled, SubscriptionMigrated, SubscriptionMigrationRejected, SubscriptionMigrationCancelled.

What this DD does NOT own

- **BPMN files** — per-operator satellites.
- **Worker implementations** — DD_SUB-WF-FRAMEWORK-01-WORKERS.
- **Subscription state machine** — SUB-LM-01.
- **Package + Package Version catalog** — SIP-01.
- **Service catalog + Provisioner interfaces** — PLM-CFG-01.
- **HomePass row mutations** — RLM-CFG-01. Read-only.
- **WO state machine, BPMN, contractor model, finalization mechanics** — WO-01-FRAMEWORK + WO-01-FLOW-SHIFTING.
- **Equipment binding mutations** — handled internally by WO-01 SHIFTING flow.
- **Migration-fee math + BillableEvent definitions + signed proration math** — BIL-01 v1.1 + BIL-CFG-01 v1.0.
- **Source-side service teardown + target-side activation + MODIFY_SERVICE fan-out** — FUL-03 v1.1 + FUL-04 v1.1.
- **HomePass bookkeeping** — FUL-07.

- **Cross-technology technical migration** — FUL-10. This DD starts and monitors the operation; FUL-10 performs the technical sequence.
- **Customer notification template + channel** — NOT-01.

Regional flexibility

Lever	Examples
BPMN file	One per operator: Subscription_Migration_<operator>.bpmn. Operator-specific branching, payment-timeout durations, notification policy live in the BPMN.
Drools package rules.subscription.migration.<operator>	Operator-specific validation thresholds, equipment-path inference matrix, disconnect-mode defaults, cycle-anchor defaults.
subscription_migration_config row	Per-operator defaults: self-service enablement, effective-timing default, cycle-anchor default, scheduled-window length, customer-notification policy.
WO-01 FLOW-SHIFTING (referenced)	Per-operator job-type codes, equipment-recovery policy, contractor-payment side effects.
FUL-10 Technology Migration Fulfillment	Per-operator cross-tech migration handling. Enables ADMIN_TECH_MIGRATION when configured for the operator.

2. Tables overview

This DD owns one table: subscription_migration_config (per-operator config).

Cross-operation audit anchor: subscription_operation (framework-owned).

Subscription master fields updated (owned by SUB-LM-01): homepass_id, previous_homepass_id, package_ref, package_version_id, previous_package_ref, previous_package_version_id, last_migrated_at, last_status_changed_at. current_transition_type = MIGRATION during PENDING_MIGRATION.

3. Schemas

3.1 subscription_migration_config

One row per operator.

```
CREATE TABLE subscription_migration_config (
  operator_code          TEXT PRIMARY KEY,
  customer_self_service_enabled  BOOLEAN NOT NULL DEFAULT FALSE,
  pay_first_required      BOOLEAN NOT NULL DEFAULT TRUE,
  default_effective_timing TEXT NOT NULL DEFAULT 'SCHEDULED_AT',
  max_future_scheduled_days INTEGER NOT NULL DEFAULT 60,
  default_disconnect_mode_keep_existing TEXT NOT NULL DEFAULT 'TECH_DISPATCH',
  default_cycle_anchor_policy TEXT NOT NULL DEFAULT 'PRESERVE',
  admin_tech_migration_enabled  BOOLEAN NOT NULL DEFAULT FALSE,
  customer_notification_enabled BOOLEAN NOT NULL DEFAULT TRUE,
  updated_at              TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_by              TEXT NOT NULL,
  CHECK (default_effective_timing IN ('IMMEDIATE', 'END_OF_CURRENT_CYCLE', 'SCHEDULED_AT')),
  CHECK (default_disconnect_mode_keep_existing IN ('TECH_DISPATCH', 'CUSTOMER_SELF_MOVE')),
  CHECK (default_cycle_anchor_policy IN ('PRESERVE', 'RESET_TO_OPERATION_DATE')),
  CHECK (max_future_scheduled_days > 0)
);
```

Column	Notes
customer_self_service_enabled	When true, CUSTOMER_REQUESTED_MIGRATION may be initiated from customer portal.
pay_first_required	When true AND BIL-01 returns positive totals.net: pay-first gate at COMMIT time.
default_effective_timing	Default applied when request omits.
max_future_scheduled_days	Max window for SCHEDULED_AT.scheduledAt.
default_disconnect_mode_keep_existing	Default for KEEP_EXISTING. NEW_INSTALL forces TECH_DISPATCH.
default_cycle_anchor_policy	Default cycle-anchor policy when request omits. Operator policy.

Column	Notes
admin_tech_migration_enabled	When true, ADMIN_TECH_MIGRATION trigger is accepted and the workflow calls FUL-10. Set to false for operators that have not approved cross-tech migration yet.
customer_notification_enabled	Notification at acceptance + commit. NOT-01 owns template/channel/language.

WO timeouts are NOT configured here — they live in WO-01's wo_kind_config.

Sample data:

operator_code	customer_self_service_enabled	pay_first_required	default_effective_timing	max_future_scheduled_days	default_disconnect_mode_key_existing	default_cycle_anchor_policy	admin_tech_migration_enabled	customer_notification_enabled
WIK	true	true	SCHEDULED_A T	60	TECH_DISPATCH	PRESERVE	false	true
WUG	false	true	SCHEDULED_A T	60	TECH_DISPATCH	PRESERVE	false	true
WTZ	false	true	SCHEDULED_A T	60	TECH_DISPATCH	PRESERVE	false	true
YASSN	false	true	SCHEDULED_A T	60	TECH_DISPATCH	PRESERVE	false	true

4. Rules

4.1 State precondition + transition

Rule	Description
R-SUB-WF-MIGRATION-01-S-1	Migration requires subscription's status_code = ACTIVE at request time AND at commit time. Other statuses return 409 INVALID_STATE_FOR_MIGRATION.
R-SUB-WF-MIGRATION-01-S-2	The transition at commit is ACTIVE → PENDING_MIGRATION → ACTIVE. PENDING_MIGRATION entered only during commit window.
R-SUB-WF-MIGRATION-01-S-3	Per framework R-FW-1, migration COMMIT is mutually exclusive with other state-affecting operations. Scheduled-future migration in wait state allows other operations (which cancel the pending migration).
R-SUB-WF-MIGRATION-01-S-4	current_transition_type = MIGRATION during PENDING_MIGRATION. current_transition_reason_code = migrationTrigger. Not set during scheduled-future wait.
R-SUB-WF-MIGRATION-01-S-5	Restrictions on the subscription preserved post-migration. New HomePass + new Package inherit the restriction set. Operator policy may decide to clear or re-evaluate; this DD does not.
R-SUB-WF-MIGRATION-01-S-6	At most one pending-future migration per subscription. Second non-IMMEDIATE returns 409 PENDING_MIGRATION_ALREADY_SCHEDULED.

4.2 Trigger taxonomy + role gating

Rule	Description
R-SUB-WF-MIGRATION-01-RT-1	2 triggers: CUSTOMER_REQUESTED_MIGRATION and ADMIN_TECH_MIGRATION. Carried as migrationTrigger. Unknown values → 400 UNKNOWN_MIGRATION_TRIGGER.
R-SUB-WF-MIGRATION-01-RT-2	Role gating: CUSTOMER_REQUESTED_MIGRATION requires CUSTOMER_SELF_SERVICE OR CALL_CENTER OR SUBSCRIPTION_OPERATOR; ADMIN_TECH_MIGRATION requires SUBSCRIPTION_ADMIN. Mismatched role returns 403.
R-SUB-WF-MIGRATION-01-RT-3	CUSTOMER_SELF_SERVICE role allowed only when subscription_migration_config.customer_self_service_enabled = true.
R-SUB-WF-MIGRATION-01-RT-4	ADMIN_TECH_MIGRATION trigger is gated by subscription_migration_config.admin_tech_migration_enabled. When false, return 400 TECHNOLOGY_MIGRATION_NOT_ENABLED. When true, call FUL-10 for the technical fulfillment sequence.

4.3 Target-Package validation

Rule	Description
R-SUB-WF-MIGRATION-01-TP-1	targetPackageRef MUST reference a sellable Package in SIP-01 within the same operator. Otherwise 400 TARGET_PACKAGE_NOT_FOUND or TARGET_PACKAGE_NOT_SELLABLE.
R-SUB-WF-MIGRATION-01-TP-2	targetPackageRef MUST differ from source package_ref. Same-Package returns 400 SAME_PACKAGE (use RELOCATION instead).
R-SUB-WF-MIGRATION-01-TP-3	Target Package's billing_mode MUST match source billing_mode. Cross-mode (PREPAID ↔ POSTPAID) returns 400 BILLING_MODE_MISMATCH (operator policy — may relax in future).
R-SUB-WF-MIGRATION-01-TP-4	Target Package's currency MUST match source currency. Cross-currency returns 400 CURRENCY_MISMATCH.
R-SUB-WF-MIGRATION-01-TP-5	The Package version is pinned at commit time. Even for SCHEDULED_AT, the version captured at commit-time revalidation is the one stored on the subscription.

4.4 Target-HomePass validation

Rule	Description
R-SUB-WF-MIGRATION-01-TH-1	targetHomepassId MUST reference a HomePass with status_code.is_sellable = true. Otherwise 400 TARGET_HOMEPASS_NOT_SELLABLE.
R-SUB-WF-MIGRATION-01-TH-2	targetHomepassId MAY equal source homepass_id (Package-only change at same HomePass — degenerate case; UPGRADE/DOWNGRADE may be a better fit).
R-SUB-WF-MIGRATION-01-TH-3	Target HomePass MUST physically support every service family required by the TARGET Package. Per Service in target package_composition, service.service_management_key MUST be present in targetHomepass.service_management_endpoints keys + the endpoint's port MUST NOT be null (commit-time enforcement). Missing keys → 400 TARGET_PACKAGE_NOT_AVAILABLE_AT_TARGET_HOMEPASS with missing keys list.
R-SUB-WF-MIGRATION-01-TH-4	If source.networkRootType != target.networkRootType AND migrationTrigger = CUSTOMER_REQUESTED_MIGRATION: reject with 400 CROSS_TECH_CUSTOMER_REQUEST_NOT_ALLOWED unless the operator explicitly enables customer-requested cross-tech migration. Cross-tech is normally admin-driven through FUL-10.

4.5 Equipment-path + equipment-delta resolution

Rule	Description
R-SUB-WF-MIGRATION-01-EQ-1	equipmentPath ∈ {KEEP_EXISTING, NEW_INSTALL}. If request supplies a value, Drools validates; otherwise auto-infers.
R-SUB-WF-MIGRATION-01-EQ-2	Auto-inference: if sourceNetworkRootType == targetNetworkRootType AND target Package's equipment requirements are a SUBSET of source Package's (i.e., no new equipment needed): inferred = KEEP_EXISTING. Otherwise inferred = NEW_INSTALL.
R-SUB-WF-MIGRATION-01-EQ-3	Request override KEEP_EXISTING with cross-tech OR with target Package requiring equipment not present at source returns 400 EQUIPMENT_INCOMPATIBLE_WITH_TARGET.
R-SUB-WF-MIGRATION-01-EQ-4	equipmentDelta is computed: addedEquipmentKinds[] (target Package needs but source doesn't) + obsoleteEquipmentKinds[] (source has but target doesn't need). Carried to WO-01 via flowContext for the SHIFTING flow's internal equipment handling.
R-SUB-WF-MIGRATION-01-EQ-5	The resolved equipmentPath + equipmentDelta are persisted to the Kafka event payload + the framework's subscription_operation audit row.

4.6 Disconnect mode resolution

Rule	Description
R-SUB-WF-MIGRATION-01-DM-1	disconnectMode ∈ {TECH_DISPATCH, CUSTOMER_SELF_MOVE}. Resolution: (a) equipmentPath = NEW_INSTALL → force TECH_DISPATCH; (b) equipmentPath = KEEP_EXISTING + request supplies → use request; (c) request omits → default from subscription_migration_config.default_disconnect_mode_keep_existing.

Rule	Description
R-SUB-WF-MIGRATION-01-DM-2	When disconnectMode = TECH_DISPATCH: workflow creates SHIFTING WO via WO-01 POST /api/work-orders (per DD_WO-01-CONTRACT-FOR-SUB-WF §1.3), parks on WorkOrderFinalized. Same integration as RELOCATION-01 v1.1.
R-SUB-WF-MIGRATION-01-DM-3	When disconnectMode = CUSTOMER_SELF_MOVE: NO SHIFTING WO is created. Customer self-handles physical move. Equipment bindings remain unchanged on subscription. Only valid for KEEP_EXISTING.
R-SUB-WF-MIGRATION-01-DM-4	Resolved disconnectMode persisted to Kafka event payload + subscription_operation audit row.

4.7 Billing

Rule	Description
R-SUB-WF-MIGRATION-01-B-1	Workflow calls bil01-emit-intent with intentCode = MIGRATION_INTENT. Context: {sourceHomepassId, sourceFranchiseId, sourceNetworkRootType, sourcePackageRef, sourcePackageVersionId, sourcePackagePrice, targetHomepassId, targetFranchiseId, targetNetworkRootType, sameFranchise, targetPackageRef, targetPackageVersionId, targetPackagePrice, equipmentPath, disconnectMode, migrationTrigger, cycleAnchorPolicy, effectiveTiming, scheduledCommitAt?, cycleWindow}.
R-SUB-WF-MIGRATION-01-B-2	BIL-01 v1.1 resolves matching BillableEvents from BIL-CFG-01 v1.0 per the cycleAnchorPolicy: PRESERVE → MIGRATION_FEE + MIGRATION_PRORATE_DELTA (signed, may be charge or credit); RESET_TO_OPERATION_DATE → MIGRATION_FEE + OLD_CYCLE_CREDIT + FIRST_CYCLE_CHARGE. Plus EQUIPMENT_REPLACEMENT_FEE if equipmentPath = NEW_INSTALL.
R-SUB-WF-MIGRATION-01-B-3	END_OF_CURRENT_CYCLE + PRESERVE: BIL-01 short-circuits proration to zero. Only MIGRATION_FEE (and possibly EQUIPMENT_REPLACEMENT_FEE) apply.
R-SUB-WF-MIGRATION-01-B-4	If pay_first_required = true AND totals.net > 0: pay-first gate at COMMIT time. Workflow waits for payment-confirmed before creating SHIFTING WO (or before proceeding past self-move acknowledgement). Payment timeout (7 days operator default) → cancel migration.
R-SUB-WF-MIGRATION-01-B-5	If totals.net <= 0 (net credit — possible with DOWNGRADE-flavored migration + RESET cycle pair where credit exceeds new charge): POSTPAID rejected with REQUIRES_CREDIT_NOTE_WORKFLOW per BIL-01 v1.1 R-SE-3 → workflow emits Rejected with BILLING_REJECTED_NEGATIVE_NET. PREPAID wallet absorbs the credit (top-up).

4.8 Cycle anchor policy

Rule	Description
R-SUB-WF-MIGRATION-01-CA-1	Request supplies cycleAnchorPolicy ∈ {PRESERVE, RESET_TO_OPERATION_DATE}. If absent, default from subscription_migration_config.default_cycle_anchor_policy. Unknown → 400 UNKNOWN_CYCLE_ANCHOR_POLICY.
R-SUB-WF-MIGRATION-01-CA-2	RESET_TO_OPERATION_DATE + effectiveTiming = END_OF_CURRENT_CYCLE: rejected with 400 INCOMPATIBLE_CYCLE_ANCHOR_AND_EFFECTIVE_TIMING (redundant — END_OF_CURRENT_CYCLE already starts new cycle on operation date).
R-SUB-WF-MIGRATION-01-CA-3	Resolved cycleAnchorPolicy persisted to Kafka event payload + subscription_operation audit row. Carried to BIL-01 v1.1 context per R-B-1.

4.9 Fulfillment sequence

Rule	Description
R-SUB-WF-MIGRATION-01-F-1	Fulfillment sequence (in order, during commit window): (a) FUL-04 SUSPEND_SERVICE at source HomePass with source Package composition; (b) FUL-07 RELEASE on source HomePass; (c) physical work — if disconnectMode = TECH_DISPATCH: create SHIFTING WO via POST /api/work-orders with flowContext={equipmentPath, disconnectMode, sourceHomepassId, targetHomepassId, equipmentDelta}, park on WorkOrderFinalized; if CUSTOMER_SELF_MOVE: log self-move acknowledgement; (d) FUL-03 ACTIVATE_SERVICE at target HomePass with TARGET Package composition + equipment bindings (from WorkOrderFinalized.finalEquipmentBindings for tech-dispatch; from pre-existing for self-move) + target service_management_endpoints; (e) FUL-03 MODIFY_SERVICE at target — diff source vs target Package composition, dispatch per-Service add/drop/modify per FUL-03 v1.1 §3; (f) FUL-07 CLAIM on target HomePass; (g) commit master fields atomically (homepass_id, previous_homepass_id, package_ref, package_version_id, previous_package_ref, previous_package_version_id, last_migrated_at).
R-SUB-WF-MIGRATION-01-F-2	Atomic-switch: subscription stays ACTIVE on source HomePass + source Package until step (f) succeeds. Failure in any step before (f) → revert: FUL-04 RESUME at source (best-effort), FUL-07 CLAIM back on source (best-effort), revert state to ACTIVE, emit SubscriptionMigrationRejected. If WO already finalized but later step failed → operational alert (equipment moved + customer expecting service).
R-SUB-WF-MIGRATION-01-F-3	WO timeout handled by WO-01 framework (wo_kind_config(kind=SHIFTING).default_timeout_days). On WorkOrderTimedOut, BPMN runs compensation per F-2 + emits Rejected with rejectionReasonCode = WO_TIMEOUT.
R-SUB-WF-MIGRATION-01-F-4	WO cancellation: dispatcher force-cancels SHIFTING WO → WorkOrderCancelled fires the BPMN's boundary; compensation runs per F-2 + emits Rejected with rejectionReasonCode = WO_CANCELLED_BY_DISPATCHER.
R-SUB-WF-MIGRATION-01-F-5	HomePass row is NOT mutated by this workflow. RLM-CFG-01 is read-only.

4.10 Effective timing

Rule	Description
R-SUB-WF-MIGRATION-01-EF-1	Request supplies effectiveTiming. If absent: default from subscription_migration_config.default_effective_timing.
R-SUB-WF-MIGRATION-01-EF-2	SCHEDULED_AT requires scheduledAt strictly future, within max_future_scheduled_days.
R-SUB-WF-MIGRATION-01-EF-3	For non-IMMEDIATE: BPMN parks on Camunda timer. subscription_operation.current_state = SCHEDULED. SubscriptionMigrationScheduled emitted. At fire, BPMN re-validates and enters commit sequence.
R-SUB-WF-MIGRATION-01-EF-4	Cancellation during wait: DELETE /pending-migration/{operationId} OR any state-affecting operation on same subscription.
R-SUB-WF-MIGRATION-01-EF-5	Commit-time revalidation re-checks all preconditions including target-HomePass + target-Package + cycle-anchor + equipment-path. Failures → rejectionStage = COMMIT_REVALIDATION.
R-SUB-WF-MIGRATION-01-EF-6	Cancellation in commit window not allowed via DELETE → 409 OPERATION_IN_COMMIT_WINDOW. Dispatcher may force-cancel SHIFTING WO via WO-01 endpoint.
R-SUB-WF-MIGRATION-01-EF-7	Pay-first gate at COMMIT time, not request time.

5. REST API

5.1 Trigger migration — scheduled (the typical customer case)

```
POST /api/subscriptions/{subscription_id}/migrate HTTP/1.1
Authorization: Bearer <jwt>
Content-Type: application/json
Idempotency-Key: migrate-sub_01HZ_OTIENO-hp_LAVI_073-pkg_INET_VOIP_IPTV-20260601
```

```
{
  "operatorCode":      "WIK",
  "migrationTrigger":  "CUSTOMER_REQUESTED_MIGRATION",
}
```

```

    "targetHomepassId": "hp_01HZ_LAVI_073",
    "targetPackageRef": "pkg_01HX_INET_VOIP_IPTV_RES",
    "equipmentPath": null,
    "disconnectMode": null,
    "cycleAnchorPolicy": "PRESERVE",
    "effectiveTiming": "SCHEDULED_AT",
    "scheduledAt": "2026-06-01T09:00:00+03:00",
    "notes": "Customer moving + upgrading to triple-play",
    "initiatingActorUserId": "cc-agent-wik-007",
    "initiatingActorRole": "CALL_CENTER",
    "initiatingWorkflowKind": "BO_UI",
    "initiatingWorkflowRef": "cc-ticket-2026-05-15-3201"
  }
}

→ 202 Accepted
{
  "operationId": "subop_01HZ_OTIENO_MIGRATE",
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_OTIENO_FIBER",
  "operationKind": "MIGRATION",
  "migrationTrigger": "CUSTOMER_REQUESTED_MIGRATION",
  "resolvedEquipmentPath": "NEW_INSTALL",
  "resolvedDisconnectMode": "TECH_DISPATCH",
  "resolvedCycleAnchorPolicy": "PRESERVE",
  "resolvedEffectiveTiming": "SCHEDULED_AT",
  "equipmentDelta": {"addedEquipmentKinds": ["STB_IPTV"], "obsoleteEquipmentKinds": []},
  "scheduledCommitAt": "2026-06-01T09:00:00+03:00",
  "bpmnProcessKey": "sub-migrate-wik",
  "bpmnProcessInstanceId": "pi_01HZ_OTIENO_MIGRATE",
  "currentState": "SCHEDULED",
  "startedAt": "2026-05-15T14:00:00+03:00"
}

```

(equipmentPath auto-inferred NEW_INSTALL because target Package adds IPTV — requires STB not present on source.
disconnectMode forced to TECH_DISPATCH because NEW_INSTALL.)

5.2 Trigger migration — ADMIN_Tech_MIGRATION

```

POST /api/subscriptions/{subscription_id}/migrate HTTP/1.1
{
  "operatorCode": "WIK",
  "migrationTrigger": "ADMIN_Tech_MIGRATION",
  "targetHomepassId": "hp_01HZ_<target>",
  "targetPackageRef": "pkg_<target>",
  "cycleAnchorPolicy": "PRESERVE",
  "effectiveTiming": "SCHEDULED_AT",
  "scheduledAt": "2026-07-01T09:00:00+03:00",
  "initiatingActorRole": "SUBSCRIPTION_ADMIN"
}

→ 202 Accepted
{
  "operationId": "op_migration_01J...",
  "status": "SCHEDULED",
  "technicalFulfillmentOwner": "FUL-10",
  "guidance": "Track operation status on SUB-WF; FUL-10 owns technical execution when the scheduled time arrives."
}

```

5.3 Errors

```

→ 400 Bad Request
{
  "code": "TARGET_PACKAGE_NOT_AVAILABLE_AT_TARGET_HOMEPASS",
  "targetPackageRef": "pkg_01HX_INET_VOIP_IPTV_RES",
  "targetHomepassId": "hp_01HZ_KIBE_010",
  "missingServiceManagementKeys": ["iptv_multicast"],
  "guidance": "Target HomePass does not support target Package's required service families. Select a"
}

→ 400 Bad Request
{
  "code": "EQUIPMENT_INCOMPATIBLE_WITH_TARGET",
  "sourceNetworkRootType": "<source-tech>",
  "targetNetworkRootType": "<target-tech>",
  "requestedEquipmentPath": "KEEP_EXISTING",
  "guidance": "Cross-tech migration requires NEW_INSTALL. Existing equipment cannot be re-homed on target net"
}

→ 400 Bad Request
{
  "code": "CROSS_Tech_CUSTOMER_REQUEST_NOT_ALLOWED",

```

```

    "guidance": "Cross-tech customer-requested migration is not enabled for this operator. Use ADMIN_TECH_MIGRATION through
}

→ 400 Bad Request
{
  "code": "INCOMPATIBLE_CYCLE_ANCHOR_AND_EFFECTIVE_TIMING",
  "guidance": "RESET_TO_OPERATION_DATE + END_OF_CURRENT_CYCLE is redundant. Use either PRESERVE + END_OF_CURRENT_CYCLE or
}

```

5.4 Cancel pending-future migration

```

DELETE /api/subscriptions/{subscription_id}/pending-migration/{operation_id} HTTP/1.1
{
  "operatorCode": "WIK",
  "cancellationReasonCode": "CUSTOMER_CHANGED_MIND",
  "initiatingActorUserId": "cc-agent-wik-007",
  "initiatingActorRole": "CALL_CENTER"
}

→ 200 OK
{
  "operationId": "subop_01HZ_OTIENO_MIGRATE",
  "cancellationReasonCode": "CUSTOMER_CHANGED_MIND",
  "cancelledAt": "2026-05-20T16:00:00+03:00",
  "subscriptionStaysOn": {"homepassId": "hp_01HZ_KAREN_017", "packageRef": "pkg_01HX_INET_100M_VOIP_RES"},
  "finalState": "CANCELLED"
}

```

In-commit-window cancel:

```

→ 409 Conflict
{
  "code": "OPERATION_IN_COMMIT_WINDOW",
  "currentState": "AWAITING_WO_FINALIZED",
  "guidance": "Migration is in the atomic-switch commit window. Dispatcher may force-cancel the SHIFTING WO via WO-01"
}

```

6. Kafka events emitted

Event	Topic	Fires on	Consumers
SubscriptionMigrationScheduled	sophix.subscription.subscription.migrationscheduled	Acceptance of non-IMMEDIATE	NOT-01, CVM, EM-04
SubscriptionMigrated	sophix.subscription.subscription.migrated	Successful commit	NOT-01, BIL-02 (cycle accrual at new HomePass + new Package), CVM (move + tier-change analytics), EM-04
SubscriptionMigrationCancelled	sophix.subscription.subscription.migrationcancelled	Pending-future cancelled	NOT-01, CVM, EM-04
SubscriptionMigrationRejected	sophix.subscription.subscription.migrationrejected	Workflow failure	NOT-01, ICN-01 ops alert, EM-04

Sample SubscriptionMigrationScheduled:

```

{
  "eventType": "SubscriptionMigrationScheduled",
  "aggregateId": "sub_01HZ_OTIENO_FIBER",
  "timestamp": "2026-05-15T14:00:01+03:00",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_OTIENO_FIBER",
    "customerId": "cust_01HZ_OTIENO",
    "operationId": "subop_01HZ_OTIENO_MIGRATE",
    "migrationTrigger": "CUSTOMER_REQUESTED_MIGRATION",
    "sourceHomepassId": "hp_01HZ_KAREN_017",
    "sourceFranchiseId": "frn_NRB_KAREN_RES",
    "sourceNetworkRootType": "<source-root-type>",
    "sourcePackageRef": "pkg_01HX_INET_100M_VOIP_RES",
    "sourcePackageVersionId": "pkv_01HX_INET_100M_V3",
    "targetHomepassId": "hp_01HZ_LAVI_073",
    "targetFranchiseId": "frn_NRB_LAVI_RES",
    "targetNetworkRootType": "<target-root-type>",
    "sameFranchise": false,
    "targetPackageRef": "pkg_01HX_INET_VOIP_IPTV_RES",
    "targetPackageVersionId": "pkv_01HX_INET_VOIP_IPTV_V1",
    "resolvedEquipmentPath": "NEW_INSTALL",
    "resolvedDisconnectMode": "TECH_DISPATCH",
    "resolvedCycleAnchorPolicy": "PRESERVE",
  }
}

```

```

    "resolvedEffectiveTiming": "SCHEDULED_AT",
    "equipmentDelta": {
      "addedEquipmentKinds": ["STB_IPTV"],
      "obsoleteEquipmentKinds": []
    },
    "scheduledCommitAt": "2026-06-01T09:00:00+03:00",
    "initiatingActorUserId": "cc-agent-wik-007",
    "initiatingActorRole": "CALL_CENTER"
  }
}

```

Sample SubscriptionMigrated:

```

{
  "eventType": "SubscriptionMigrated",
  "aggregateId": "sub_01HZ_OTIENO_FIBER",
  "timestamp": "2026-06-01T10:55:42+03:00",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_OTIENO_FIBER",
    "operationId": "subop_01HZ_OTIENO_MIGRATE",
    "migrationTrigger": "CUSTOMER_REQUESTED_MIGRATION",
    "sourceHomepassId": "hp_01HZ_KAREN_017",
    "targetHomepassId": "hp_01HZ_LAVI_073",
    "sameFranchise": false,
    "sourcePackageRef": "pkg_01HX_INET_100M_VOIP_RES",
    "targetPackageRef": "pkg_01HX_INET_VOIP_IPTV_RES",
    "resolvedEquipmentPath": "NEW_INSTALL",
    "resolvedDisconnectMode": "TECH_DISPATCH",
    "resolvedCycleAnchorPolicy": "PRESERVE",
    "resolvedEffectiveTiming": "SCHEDULED_AT",
    "scheduledCommitAt": "2026-06-01T09:00:00+03:00",
    "actualCommitAt": "2026-06-01T10:55:42+03:00",
    "shiftingWoId": "wo_01HZ_OTIENO_SHIFT_5501",
    "migrationFeesNet": 6500.00,
    "currency": "KES",
    "billingDetails": {
      "lineItems": [
        {
          "code": "MIGRATION_FEE",
          "amount": 1500.00
        },
        {
          "code": "MIGRATION_PRORATE_DELTA",
          "amount": 5000.00,
          "sign": "POSITIVE"
        }
      ],
      "totalsNet": 7540.00
    },
    "initiatingActorUserId": "cc-agent-wik-007",
    "initiatingActorRole": "CALL_CENTER"
  }
}

```

7. Cross-DD coordination

7.1 Upstream callers

Caller	migrationTrigger	Role
Customer portal	CUSTOMER_REQUESTED_MIGRATION	CUSTOMER_SELF_SERVICE (subject to config)
Call Center	CUSTOMER_REQUESTED_MIGRATION	CALL_CENTER
BO admin	ADMIN_Tech_MIGRATION	SUBSCRIPTION_ADMIN

7.2 Sibling modules

Module	Direction	Contract
SUB-WF-FRAMEWORK-01 (parent)	This DD composes framework primitives	subscription_operation row + BPMN start
SUB-WF-FRAMEWORK-01-WORKERS	This DD's BPMNs reference workers by topic	Standard framework workers + wo-create-shifting + wo-cancel
SUB-LM-01	Workers write status + master fields atomically	homepass_id, previous_homepass_id, package_ref, package_version_id, previous_package_ref, previous_package_version_id, last_migrated_at
RLM-CFG-01	Workers read source + target HomePass	service_management_endpoints, status_code, network_path root node type, franchise
SIP-01	Workers read source + target Packages	Package composition, version, currency, billing mode

Module	Direction	Contract
PLM-CFG-01 v1.0	Workers read service definitions	service_management_key, equipment requirements, RestrictionCapableProvisioner interface
BIL-01 v1.1	bil01-emit-intent worker	MIGRATION_INTENT — fee + signed proration delta + cycle-anchor RESET pair
BIL-CFG-01 v1.0	BillableEvent catalog	MIGRATION_FEE, MIGRATION_PRORATE_DELTA (SIGNED), OLD_CYCLE_CREDIT, FIRST_CYCLE_CHARGE, EQUIPMENT_REPLACEMENT_FEE
FUL-04 v1.1	ful-call-suspend worker	SUSPEND_SERVICE at source HomePass with reasonCategory=SOURCE_DECOMMISSION
FUL-03 v1.1	ful-call-activate + ful-call-modify-service workers	ACTIVATE_SERVICE + MODIFY_SERVICE at target HomePass
FUL-07 v1.0	ful-call-release-homepass + ful-call-claim-homepass	HomePass bookkeeping
WO-01 (FRAMEWORK + FLOW-SHIFTING)	wo-create-shifting worker (when same-tech disconnectMode=TECH_DISPATCH)	Per DD_WO-01-CONTRACT-FOR-SUB-WF — POST /api/work-orders with kind=SHIFTING
FUL-10 Technology Migration Fulfillment	ful10-start-technology-migration worker (when cross-tech ADMIN_TECH_MIGRATION is enabled)	Owns technical cross-tech sequence and calls WO/OSR/PROV-INT as needed.
NOT-01	notification-send worker	Customer notification
BIL-02 / CVM / EM-04	Consume Kafka events	Downstream reporting

7.3 Build order

This DD deploys after:

1. SUB-WF-FRAMEWORK-01 v1.0
2. SUB-LM-01 v1.0
3. SIP-01 v1.0, PLM-CFG-01 v1.0, RLM-CFG-01 v1.0
4. FUL-03 v1.1, FUL-04 v1.1, FUL-07 v1.0
5. BIL-01 v1.1, BIL-CFG-01 v1.0
6. WO-01 framework + FLOW-SHIFTING v1.0
7. DD_WO-01-CONTRACT-FOR-SUB-WF v1.0
8. SUB-WF-RELOCATION-01 v1.1 (sister workflow; shares many integration patterns)
9. At least one FLOW satellite per operator (FLOW-WIK v1.1 ships alongside this parent)

7.4 Stub debt + open coordination items

- **FUL-10 operator enablement** — ADMIN_TECH_MIGRATION is available only for operators with FUL-10 configuration, test data, and rollback procedures approved.
- **WUG / WTZ / YASSN FLOW satellites** — pending operator deep dives.
- **Operator-specific job-type code mapping** — operator's WO-01 Drools KJAR handles (kind=SHIFTING, sourceNetworkRootType, equipmentPath, equipmentDelta) → operator job-type code.

B — Current TZ

Status: To be filled in after codebase cartography review.